

## **Animacy, intention, and null objects in English**

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### **Abstract**

It has long been known that the objects of at least some English verbs that are normally transitive can be omitted in at least some circumstances. In recent work, the animacy of the subject has emerged as a potential factor influencing sentence acceptability. While different theories make different predictions about where and how such effects emerge, little quantitative research exists on the extent and nature of animacy-related effects upon object omission. This paper presents a series of three experiments to elicit acceptability judgements on sentences differing in subject animacy and in the presence or absence of an overt object. It was found that not only are animacy and object presence significant factors in their own right, but a significant interaction between animacy and object presence also exists for at least some sentence pairs. It was also found that a similar degradation of object omission exists for animate subjects not in control of their actions, as opposed to animate subjects in control. This suggests that a key difference between animate and inanimate subjects may be the former's intentional status.

**Keywords:** animacy, intentionality, object drop, syntax, grammaticality

## 1 Introduction

It is well known that the objects of at least some English verbs that are normally transitive can be omitted in at least some circumstances, as in (1).

- (1) a. I've already eaten  
 b. When the car broke down, Mary pushed and John steered

This phenomenon is distinct from the more pervasive object-drop phenomena seen in languages such as Brazilian Portuguese (e.g. Farrell 1990; Cyrino 2021) and Japanese (e.g. Saito 2007), where almost any object can be omitted, subject to constraints such as recoverability and/or topicality. In English, such factors are not sufficient to allow object omission, as seen in (2).

(2) (from Cyrino 2021: 400)

- a. a estudante levou o livro para a biblioteca  
 the-F.SG student take-3SG.PST the-M.SG book to the-F.SG library  
 depois que ela leu  
 after that she read-3SG.PST  
 'The student took the book to the library after she read (it)'  
 b. The student took the book the the library after she read \*(it)

It has been known at least since the work of Raposo (1986) that this type of phenomenon is fundamentally different from object omission in English; unlike the latter, it is analysed in the literature as involving either empty *pro* (e.g. Farrell 1990) or ellipsis (e.g. Duguine 2014), . Moreover, Fillmore's foundational work on object omission (1969, 1986) shows that in English, it is subject to additional constraints, as seen in examples such as (3); here *eat* can occur without an object, in what Fillmore calls the Indefinite Null Complement construction, while *devour* cannot.

(3) (after Rice 1988)

- a. John ate  
 b. \*John devoured

Fillmore also discusses another type of object omission, which he terms Definite Null Complement, involving a contextually recoverable object, as in (1b).<sup>1</sup> Fillmore's conclusion was that object omission was specified in the lexicon on a verb-by-verb basis. However, extensive previous work has shown that even in English, where object omission is not

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<sup>1</sup> Other authors (e.g. Glass 2022) view "indefinite" and "definite" constructions not as discrete types but as points on a continuum.

facilitated by syntactic entities such as empty pronominal elements, the availability of object drop or object omission is not purely lexical, but rather linked to a number of factors, including telicity and modality (cf. Goldberg 2001). Yankes (2021), in his summary of the previous literature, identifies as many as seven features to which the possibility of object omission has been ascribed.

The aim of the present study is not to provide a comprehensive model of object omission in English, but rather to focus on relevant properties of the subject. In recent literature, the observation has emerged that object omission may be dispreferred with inanimate subjects (e.g. Martí 2015; Rappaport Hovav & Levin 2024; Folli & Harley forthcoming).

(4) (cf. Rappaport Hovav & Levin 2024: 11–12)

- a. Danny swept the floor
- b. Danny swept
- c. The tsunami swept the debris off the beach
- d. #The tsunami swept

(5) (cf. Folli & Harley forthcoming)

- a. After the party, John washed the dishes
- b. After the party, John washed
- c. After the drought, the rain washed the sidewalk
- d. #After the drought, the rain washed

While such contrasts seem real, little quantitative research exists to establish the extent and distribution of animacy-related effects on object omission. This paper presents the results of three experiments to evaluate the effect of subject animacy on grammatical acceptability judgments. It is found that the animacy of the subject does affect the acceptability of sentences involving object omission; it is also found that animate subjects not in control of their actions pattern closer to inanimate subjects than to animate subjects acting intentionally. To contextualize these findings, it may be useful to begin with a more detailed review of previous analyses of object omission, before proceeding to a discussion of the present study's methodology and the results obtained. It will be seen that although the findings support the study's hypotheses, a number of questions remain to be addressed.

## 2 Previous accounts

Previous research on object omission has focused on a wide range of topics. Debate has often centred around questions such as whether the omitted object is still present in some way or whether it lacks any syntactic representation. For example, some accounts analyse object omission as involving an object which has no overt realization but is still present in the syntax (e.g. Dayal 2003; Martí 2015). In other accounts, the object may be completely absent from the syntax, and its interpretation is achieved through a process of pragmatic free enrichment (e.g. Recanati 2002), the semantics of a separate lexical entry (e.g. Rappaport Hovav & Levin 2024), or the coercion of an unsaturated predicate (Folli & Harley forthcoming). Another topic explored by previous work is whether variation in object omission among different verbs is encoded in the lexicon (e.g. Fodor & Fodor 1980, Rappaport Hovav & Levin 2024) or reflects phenomena pervading the grammar as a whole (e.g. Rice 1988, Folli & Harley forthcoming). Although many of these authors have provided no explicit discussion of animacy or intention as a potential factor, it will be seen in Section 5 that a fuller understanding of such effects can be relevant to these broader theoretical questions. As a closer examination of previous work will show, a growing awareness of the relevance of animacy/intention is the result of a convergence from multiple perspectives.

### 2.1 Animacy and noun incorporation

One of the first authors to suggest the relevance of subject animacy to object omission was Martí (2015). Martí analysed object omission in English as involving the incorporation of a null noun. Drawing upon data from Dyk (1997), she observed that in Frisian, both overt noun incorporation and (English-style) object omission share a number of distinguishing properties, one of which is the requirement of animate, volitional subjects, as shown in (6)–(7).<sup>2</sup>

(6) (from Dyk 1997: 97–109)

- |    |                                      |       |                    |     |          |  |
|----|--------------------------------------|-------|--------------------|-----|----------|--|
| a. | It                                   | bern  | kleuret            | it  | plaatsje |  |
|    | the                                  | child | colour-3SG         | the | picture  |  |
|    | ‘The child is colouring the picture’ |       |                    |     |          |  |
|    |                                      |       |                    |     |          |  |
| b. | It                                   | bern  | plaatsjekleuret    |     |          |  |
|    | the                                  | child | picture.colour-3SG |     |          |  |
|    | ‘The child is colouring pictures’    |       |                    |     |          |  |

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<sup>2</sup> It should be noted, however, that these examples involve two distinct senses of the polysemous verb *kleure*, namely ‘apply pigment to’ and ‘change the apparent hue of’.

- c. It      bern    kleuret  
the    child   colour-3SG  
'The child is colouring'
- (7) a. De      ûndergeande              sinne   kleuret              it      hûs  
the    go.under-PRS.PPL      sun    colour-3SG      the    house  
'The setting sun is colouring the house'
- b. \*De      ûndergeande              sinne   hûskleuret  
the    go.under-PRS.PPL      sun    house.colour-3SG  
'The setting sun is colouring houses'
- c. \*De      ûndergeande              sinne   kleuret  
the    go.under-PRS.PPL      sun    colour-3SG  
'The setting sun is colouring'

Martí argues that object omission in Frisian is noun incorporation, involving a null indefinite N, and that is this also the case in English for both the definite and indefinite types of object omission, given that the distinguishing properties of these two constructions in Frisian are also shared by English object omission (e.g. number neutrality, narrow scope, indefinite interpretation of the incorporated N). Martí also draws explicit attention to the status of volition/intention as a key characteristic of animate subjects. Under her analysis, the association between object omission and animacy would thus result from cross-linguistic constraints on noun incorporation, although she does not attempt to provide a detailed model of how so specific a constraint might arise. While she predicts that the distribution of object omission in English and Frisian would be parallel, the testing of such a prediction is outside the scope of her work; any such cross-linguistic comparison would also require more data on the Frisian phenomena than provided by Dyk (1997).

## 2.2 Animacy and the lexicon

The idea that object omission results from lexically encoded properties has a long history (e.g. Fodor & Fodor 1980; Fillmore 1986). Fodor and Fodor (1980) argue that the potential for a transitive verb like *eat* to occur without an object is marked in the lexicon, and that, *contra* Martí (2015), no syntactic argument is present. Similarly, Fillmore (1986) argues that English lacks grammatical object drop, whilst cases of omission of the object are pragmatic and/or lexical in nature. Perhaps the first analysis to link animacy explicitly with such a lexical approach to object omission structures is that of Rappaport Hovav & Levin (2024). In this paper they investigated the hypothesis that a single-sense approach to verb alternation

may be appropriate in some cases of alternations but not in all. Their work focused on one verb, *sweep*, and identified animacy restrictions such as those in (4) and (8)–(10).

- (8) (after Rappaport Hovav & Levin 2024: 10)
- a. The frigid waves swept the deck
  - b. \*The frigid waves swept
- (9) a. The north wind swept the open tundra
- b. \*The north wind swept
- (10) a. The branch of the tree swept the window
- b. \*The branch of the tree swept

To explain such phenomena, they proposed a polysemous account of the verb, according to which “basic-*sweep*” and “broom-*sweep*” are essentially separate verbs with different meanings and different argument structures, deriving from a common root ‘*sweep*’. “Basic-*sweep*” means “x moves across a surface y while x imparts a force to y through contact”, and is commonly found with agentive and non-agentive subjects (in fact, according to their corpus, more commonly with the latter) with the object typically referring to a surface. This lexical item does not allow object omission with either agentive or non-agentive subjects: omitting the argument which refers to the surface typically leads to ungrammaticality. The second sense of *sweep* is “broom-*sweep*”, derived from the first one via conventionalization to give a narrower sense of the verb, which is restricted to animate subjects who are responsible for using a broom to act on a surface. This would result in a separate lexical entry stipulating both a specifically agentive role for the subject and the optionality of the object. They also highlight the necessarily intentional status of the subject of “broom-*sweep*” (2024: 7). Their account thus predicts that the permissible subjects for “broom-*sweep*” would thus parallel those for obligatorily agentive verbs such as *murder* (2024: 3).

### 2.3 Animacy and sortal criteria

A more extensive exploration of the role of animacy and object omission is provided by Folli & Harley (forthcoming). In this paper they consider verbs of contact, as in (11)–(12) below:

- (11) a. After the party, John washed the dishes
- b. After the drought, the rain washed the sidewalk
- (12) a. John pushed the car to the kerb
- b. The glacier pushed the boulder into the sea

They notice that in the frame without an object, only agentive subjects are possible:

- (13) a. After the party, John washed
- b. #After the drought, the rain washed

- (14) a. At the signal, John pushed  
 b. #As soon as the temperature reached 80°, the glacier pushed

In line with prior work such as Mittwoch (1982, 2005) and Glass (2014), they show that the availability of object omission is pervasive and that the animacy effect they are investigating is orthogonal to Filmore's distinction between definite and indefinite null objects. They also show that an account in terms of verb frame alternation as in Glass (2022) is not tenable and that an incorporation account which uses existential closure to derive the interpretation of the objectless form is equally problematic. Instead, they develop an analysis which presents object omission as the result of a productive syntactic phenomenon, whereby transitive verbs can be coerced into intransitive syntactic frames (cf. Borer 2005). In order for the resulting predicates to be interpretable, an agentivity restriction is introduced to provide the needed sortal criteria and thus to allow the utterance's truth conditions to be evaluated. This account predicts that only animate subjects would be permissible with object omission, as only subjects of this sort are compatible with the criteria. In this analysis too, acceptability depends not only on animacy but crucially also on intentionality.

## 2.4 Animacy and conventionalization

A large part of the prior literature on this topic has connected the availability of object omission to a variety of semantic/pragmatic/discourse features which facilitate the omission of the object with a given verb, particularly in indefinite cases. In particular, several accounts of object omission relate its acceptability to the routine, conventionalized nature and the prototypicality of the event denoted. For example, according to Goldberg (2001), object omission is licensed in constructions in which prototypical scenes are evoked and therefore the object is low in prominence. Similarly, Glass proposes that object omission is facilitated by the extent to which the verb describes 'a series of actions that are well-known and conventional within a community' (2022: 53), and in particular by the routinization of these actions. Unlike Rappaport Hovav & Levin (2024), who present conventionalization and animacy as separate but interrelated factors, these authors seldom make explicit reference to animacy as a potential factor; however, the examples of object omission that they cite generally involve animate, usually intentional subjects, as in (15)–(16).

(15) (Goldberg 2001: 506)

- a. Tigers only kill at night  
 b. Pat gave and gave, but Chris just took and took

(16) (Glass 2014: 122)

- a. Officers pursue because they believe the person they are trying to stop must

present a serious threat

- b. We are all human and we're always discovering

If conventionalization is a factor, it is possible that conventionalized routines involving inanimate subjects might be less frequent, perhaps because of their more restricted range of capabilities; in this case an association between animacy and object omission might arise epiphenomenally. However, it is also possible that conventionalized routines involving inanimate subjects are common enough to prevent any such association. As the authors mentioned above provide little explicit discussion of animacy,<sup>3</sup> they make no predictions on this point.

## **2.5 Role of the present study**

It can be seen that all these accounts are converging, from multiple perspectives, upon the same prediction, namely that differences in the animacy and/or intentionality of the subject will produce differences in the acceptability of object omission. However, the evidence base for such a prediction has been relatively small. Most previous work on object omission has relied solely on introspective judgements, involving a relatively small number of sentences, while the corpus-based and experimental studies of Glass (2022) did not include animacy or intentionality as specific factors for analysis. The goal of the present study is to remedy this lack of data.

The primary research question of this study is whether animacy and/or intentionality exercise a significant effect upon the acceptability of object omission. More specifically, it tests two hypotheses: that acceptability judgement ratings will show an interaction between object omission and animacy as factors, and that a similar interaction will exist between object omission and intentionality. These hypotheses are in themselves compatible with all the theories of object omission described above. The null hypothesis is that no such interactions would be found. Our prediction is that interactions will be found; however, no prediction is made as to whether object omission, animacy, or intentionality will be significant as main effects, given that both significance and non-significance would be equally compatible with all theories. There are other respects in which the theories described above do generate different predictions; these differences will be discussed in Section 6, following the discussion of the present study's methodology and results.

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<sup>3</sup> Goldberg (2001: 520–521) does discuss intentionality in a different context, that of caused motion constructions with overt objects.

### 3 Experiment 1

Experiment 1, like the subsequent experiments, consisted of a grammaticality judgement survey, administered online to adult participants. The design of this experiment was preregistered on OSF prior to data collection.<sup>4</sup> In this section, details of the materials, procedures and experimental design will be followed by details of the specific hypothesis tested and how the results confirm this hypothesis.

#### 3.1 Participants

Participants were 80 adults who were residents of the UK, 18 years of age or older, and who reported English as their first language. An additional 12 participants were excluded for answering too many control questions incorrectly (see Section 3.2). Participant recruitment was conducted online using Prolific (<https://www.prolific.com>). Ethical approval for this experiment was granted by [ANONYMIZED]. Informed consent was required prior to participation. Each participant received compensation of US\$10 per hour for a 10-minute study.

#### 3.2 Materials and procedure

In Experiment 1, as in the subsequent experiments, each stimulus consisted of a pair of sentences. A context sentence, which introduced the object into the discourse, was followed by a test sentence, in which the presence or absence of an overt object was manipulated, as in (17).

- (17)a. John is cleaning the windows. He isn't cleaning them very well.  
(animate subject, overt object)
- b. John is cleaning the windows. He isn't cleaning very well.  
(animate subject, no object)
- c. The rain is cleaning the windows. It isn't cleaning them very well.  
(inanimate subject, overt object)
- d. The rain is cleaning the windows. It isn't cleaning very well.  
(inanimate subject, no object)

The stimuli were constructed in as consistent a manner as possible, in order to control for factors that have been identified in previous literature as potentially relevant to object omission. For example, as aspect has been shown to influence the acceptability of object omission (e.g. Mittwoch 2005), all test sentences were constructed in the progressive. Likewise, all stimuli involved only “definite” object omission, in the sense that omitted

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<sup>4</sup> [https://osf.io/8as3k/overview?view\\_only=4870ffda455c43688aa21e7b5387e74c](https://osf.io/8as3k/overview?view_only=4870ffda455c43688aa21e7b5387e74c)

objects are specific entities already present in the discourse. As discussed in Section 2, multiple theories predict the acceptability of object omission to be degraded with inanimate subjects. This means that (17d) is predicted to receive lower ratings relative to (17c) than (17b) to (17a). None of the theories described here makes explicit predictions about the relative acceptability of (17a) and (17b), or (17a) and (17c).

Experiment 1 included both test verbs and control verbs (Table 1). While test verbs were predicted to allow object omission in at least some contexts, control verbs were selected as requiring or disallowing objects consistently in all contexts.<sup>5</sup>

| Verb                | Type                        |
|---------------------|-----------------------------|
| catch               | Test verb                   |
| clean               | Test verb                   |
| cook                | Test verb                   |
| eat                 | Test verb                   |
| paint               | Test verb                   |
| pull-1              | Test verb                   |
| pull-2 <sup>6</sup> | Test verb                   |
| push                | Test verb                   |
| squish              | Test verb                   |
| sweep               | Test verb                   |
| wash                | Test verb                   |
| wipe                | Test verb                   |
| bring               | Control verb (transitive)   |
| hold                | Control verb (transitive)   |
| make                | Control verb (transitive)   |
| remove              | Control verb (transitive)   |
| come                | Control verb (intransitive) |
| fall                | Control verb (intransitive) |
| go                  | Control verb (intransitive) |
| rest                | Control verb (intransitive) |

**Table 1:** Verbs used in Experiment 1

These controls were designed as attention checks, to identify participants who might be giving constant or random responses regardless of grammaticality. Obligatorily transitive verbs, as in (18), should not be accepted without an overt object, even in the animate condition; likewise, intransitive verbs, as in (19), should never be accepted with an overt object:

<sup>5</sup> Although *rest* can be used transitively and *remove* intransitively, the experiment used sentences constructed so as to exclude these readings (see Supplementary Files).

<sup>6</sup> In an earlier pilot phase, the verb *rewrite* was also used. *Rewrite* was found to be generally unacceptable with object omission even for animate subjects, and so it was replaced with a different example involving *pull*. In the subsequent analysis, the two *pull* examples were treated as separate and independent. For reasons of space, pilot data are not presented or analysed here.

- (18)a. John is bringing Mary flowers. He's bringing them because it's her birthday.  
 b. John is bringing Mary flowers. \*He's bringing because it's her birthday.  
 c. The wind is bringing rain. It's been bringing it all month long.  
 d. The wind is bringing rain. \*It's been bringing all month long.
- (19)a. John was carrying some books. \*He tripped and fell them on the floor.  
 b. John was carrying some books. He tripped and fell on the floor.  
 c. This trolley was carrying some cups. \*It collapsed and fell them on the floor.  
 d. This trolley was carrying some cups. It collapsed and fell on the floor.

Although all test sentences were in the progressive, control sentences such as those shown in (19) used a broader range of aspects, since their acceptability depends on factors unrelated to aspect. Each participant saw 16 control items, in addition to the 24 test items (see Section 3.3), for a total of 40 items. Participants were excluded if they gave ratings over the midpoint of 4 to at least one ungrammatical sentence for at least two control verbs.

Like the subsequent experiments, Experiment 1 was an online experiment designed using jsPsych (de Leeuw et al. 2023). After viewing and submitting the consent form, participants were taken to a training section where they were introduced to the study's Likert-style response format, in which they were asked to rate each test sentence on a scale from 1 (least acceptable) to 7 (most acceptable). They were presented with grammatical and ungrammatical test sentences and asked to rate them. Feedback was given for incorrect ratings ( $\leq 4$  for grammatical sentences and  $\geq 4$  for ungrammatical sentences), and participants were able to proceed only after giving correct ratings. The main experiment then began; participants were shown sentence pairs of a context followed by the test sentence, and asked to rate the test sentence (described in the instructions as the 'second' sentence).

### 3.3 Design

In this experiment, as in the others, the independent variables were animacy (within subjects) and object presence (between subjects); in other words, each participant viewed sentences all of which either had or lacked overt objects, and saw one animate and one inanimate sentence for each verb. The dependent variable was the sentence rating; this was provided as a Likert acceptability rating from 1 to 7, which was then standardized (z-scored) for the purposes of analysis (see Section 3.5). This design is illustrated in Table 2.

|                  | Object                  | No Object                  |
|------------------|-------------------------|----------------------------|
| <b>Animate</b>   | Animate, object (17a)   | Animate, no object (17b)   |
| <b>Inanimate</b> | Inanimate, object (17c) | Inanimate, no object (17d) |

**Table 2:** Experiment 1 design, with object/no object (between subject) and animate/inanimate (within subject)

The stimuli were presented in blocks, with blocks of four animate stimuli alternating with blocks of four inanimate stimuli. The order of verbs within the animate and inanimate sets was randomized, with test and control verbs randomized together.

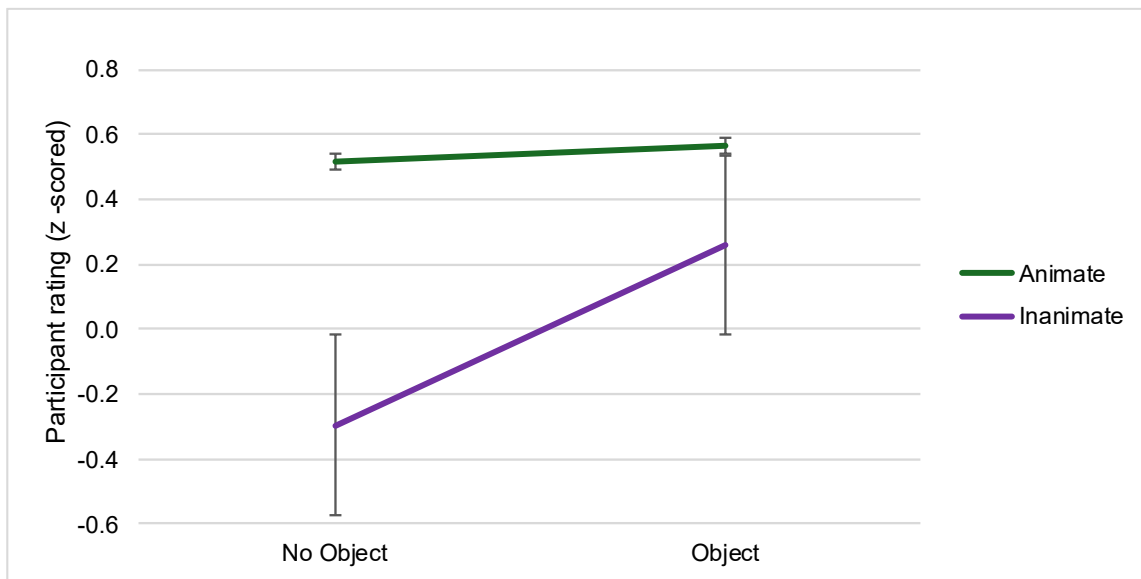
### **3.4 Predictions**

As described above, the goal of this study was to test the prediction, made by multiple theories, that animacy exercises a significant effect upon the acceptability of object omission. In terms of the design of this experiment, what is predicted is that there will be a difference in the acceptability of sentences with an object from that of sentences with object omission, depending on animacy: with an animate subject, there should be less difference between the acceptability of the two conditions than with an inanimate subject, where the no-object condition should be significantly less acceptable, resulting in an interaction between the factors of animacy and object presence.

The null hypothesis is that no such interaction will be found; i.e. that there will be no significant difference between animacy conditions in the relative acceptability of sentences with and without overt objects. No prediction was made as to whether animacy and object presence would be significant in their own right as main effects, since this is a question beyond the scope of the theories discussed in Section 2. It will be seen that the results of this experiment confirm this prediction: as described in the following section, the expected interaction is observed.

### **3.5 Analysis**

The results of Experiment 1 can be seen in Figure 1. As expected, the ratings for the condition with an inanimate subject and no overt object are lower than in any other condition. Likewise, ratings for object omission with inanimate subjects are lower relative to corresponding sentences with objects than ratings for object omission with animate subjects. These results support the predictions made regarding the relative acceptability of the different conditions.



**Figure 1:** Participant ratings by condition (Experiment 1)

To analyse these results, regression analyses were conducted upon the data using R Statistical Software (R Core Team 2023). The fixed effects were animacy and object presence, dummy coded as the Boolean values 0 (false) and 1 (true). The categorical variables of item and participant were included as random effects. The dependent variable was the z-scored Likert acceptability judgement.<sup>7</sup> A linear mixed-effects regression was conducted using the ‘lmerTest’ package (Kuznetsova et al. 2017).<sup>8</sup> Regression formulae and other analytical code are available in the Supplementary Files.

The regression provided support for the experiment’s hypotheses. In the mixed-effects linear regression, animacy was significant as a main effect ( $\beta=0.81$ ,  $t=6.20$ ,  $p<0.001$ ), as was object presence ( $\beta=0.56$ ,  $t=5.01$ ,  $p<0.001$ ); the interaction between the two was also significant ( $\beta=-0.51$ ,  $t=-3.99$ ,  $p=0.001$ ). This interaction can be attributed to a greater effect of animacy in the no-object condition than in the object condition. The results support the predictions made by the theories discussed in Section 2, and are inconsistent with the null hypothesis that no interaction between animacy and object presence would exist.

<sup>7</sup> Z-scoring was conducted to correct for differences in participant response styles (e.g. extreme vs. centered); participant ID was retained as a random effect in the model to identify any genuine participant differences in grammaticality judgements that might remain after correction for response-style differences. Without Z-scoring, the presence of participants with identical perceptions of grammaticality but different response styles would be sufficient in itself to produce significant participant-level effects.

<sup>8</sup> To ensure greater robustness (cf. Taylor et al. 2023), a cumulative link mixed-effect ordinal regression was also conducted using the ‘ordinal’ package (Christensen 2023), using the raw Likert scores as the dependent variable. The same factors and interactions were significant in both types of regression, here and in subsequent experiments; the cumulative link mixed model data are available in Appendix 1.

### 3.6 Discussion

The aim of Experiment 1 was to provide initial support for the prediction that an interaction would exist between subject animacy and object omission. However, this experiment had a relatively simple design; in particular, the fact that each verb was represented only by a single animate/inanimate stimulus pair hindered the identification of any variation that might exist at the lexeme level. The next experiment was designed with the aim of providing a more comprehensive picture, by replicating the results of the first experiment using multiple items for each verb.

## 4 Experiment 2

In order to provide more robust support for hypotheses regarding the role of animacy in object omission, a more complex design was needed, including multiple stimuli per verb and a greater number of controls. Experiment 2 was therefore designed as an extension of the grammaticality judgement tasks used in Experiment 1. As before, this experiment was preregistered prior to data collection.<sup>9</sup> It employed a similar online format and was procedurally similar in many respects. However, it will be seen below that a number of design changes were also made.

### 4.1 Participants

Experiment 2 used a larger sample size than Experiment 1, of 180 participants. This target was set using power calculations based on effect sizes from Experiment 1, using simulated random data (Vasishth et al. 2020); simulation code is available in the Supplementary Files. From the simulations it was found that this sample size would typically provide a power between 0.7 and 0.8; as substantially higher powers were found only at much greater sample sizes, this size was chosen to provide a balance between power and feasibility. Experiment 1 participants were not eligible for Experiment 2. A total of 189 participants were recruited, with 9 excluded; 7 performed poorly on control questions and 2 chose to withdraw after completing the study. Participant recruitment was again conducted online using Prolific, with the same compensation rates as before.

### 4.2 Materials and procedure

In Experiment 2, test sentences and procedures followed the same format as in the previous experiment. However, four different stimulus sets were constructed for each verb; accordingly, a smaller list of verbs was used (see Table 3). The format and nature of control items was the same as in the previous experiment; however, the design of this experiment

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<sup>9</sup> [https://osf.io/ang9y/overview?view\\_only=2077e533c6fd4fdf923d2d899cec402e](https://osf.io/ang9y/overview?view_only=2077e533c6fd4fdf923d2d899cec402e)

(see Section 4.3), in which each participant would see a greater number of ungrammatical sentences, led to a change in exclusion criteria; participants were excluded if they achieved less than 80% accuracy on all control sentences combined. As Table 3 shows, there were 6 test verbs and 4 control verbs.

| Verb   | Type                        |
|--------|-----------------------------|
| Clean  | Test verb                   |
| Cook   | Test verb                   |
| Pull   | Test verb                   |
| Push   | Test verb                   |
| Scrape | Test verb                   |
| Wipe   | Test verb                   |
| Bring  | Control verb (transitive)   |
| Make   | Control verb (transitive)   |
| Come   | Control verb (intransitive) |
| Go     | Control verb (intransitive) |

**Table 3:** Verbs used in Experiment 2

### 4.3 Design

In Experiment 2, the design was substantially different from that of the previous experiment. The variables involved were the same, with sentence rating as the dependent variable and with animacy and object presence as independent variables. As before, each participant was assigned to an object condition, so that all stimuli would either have or lack overt objects.<sup>10</sup> Each participant was presented with 4 scenarios per test verb, 2 in each animacy condition. For the control verbs that were always grammatical in their object condition, participants were presented with 4 scenarios per verb in the same way; for the control verbs that were always ungrammatical in that condition, subjects received 8 scenarios (4 scenarios  $\times$  2 animacy conditions), to provide a sufficient number of exclusion items. This gave a total of 48 stimuli, which were presented with control and test items alternating. A Latin-square design was used in the random assignment of participants to stimuli sets (see Table 4). To sensitize participants to the type of judgements involved, the first 8 control sentences were always ungrammatical. However, the order of these sentences was randomly shuffled for each participant. The test sentences were divided into two blocks, so that half the sentences for each verb occurred within the first half of the study and the others within the second half. The

<sup>10</sup> Design considerations proved to be important in the elicitation of these judgements. A parallel experiment ([https://osf.io/akqcr/overview?view\\_only=24a8c67c6a6b47b583545b5ca03c638f](https://osf.io/akqcr/overview?view_only=24a8c67c6a6b47b583545b5ca03c638f)) was performed with the same questions and protocol, but a within-subjects design (2 animacy conditions  $\times$  2 object conditions for each verb), and the interaction between effects was nonsignificant. The greater variety of factors may have reduced participants' sensitivity to the relevant distinctions.

order of sentences within each block was randomly shuffled, as were the remaining control sentences.

|        | <b>Group A</b>                                                                           | <b>Group B</b>                                                                      | <b>Group C</b>                                                                         | <b>Group D</b>                                                                    |
|--------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Item 1 | Mary is pushing some toy boats across the pond. She's pushing them almost silently.      | Mary is pushing some toy boats across the pond. She's pushing almost silently.      | The wind is pushing some toy boats across the pond. It's pushing them almost silently. | The wind is pushing some toy boats across the pond. It's pushing almost silently. |
| Item 2 | The wind is pushing some empty boxes across the street. It's pushing them quite noisily. | The wind is pushing some empty boxes across the street. It's pushing quite noisily. | Mary is pushing some empty boxes across the street. She's pushing them quite noisily.  | Mary is pushing some empty boxes across the street. She's pushing quite noisily.  |
| Item 3 | Mary is pushing the doors open. She's pushing them very softly.                          | Mary is pushing the doors open. She's pushing very softly.                          | The breeze is pushing the doors open. It's pushing them very softly.                   | The breeze is pushing the doors open. It's pushing very softly.                   |
| Item 4 | The current is pushing two canoes towards the shore. It's pushing them very slowly.      | The current is pushing two canoes towards the shore. It's pushing very slowly.      | Mary is pushing two canoes towards the shore. She's pushing them very slowly.          | Mary is pushing two canoes towards the shore. She's pushing very slowly.          |

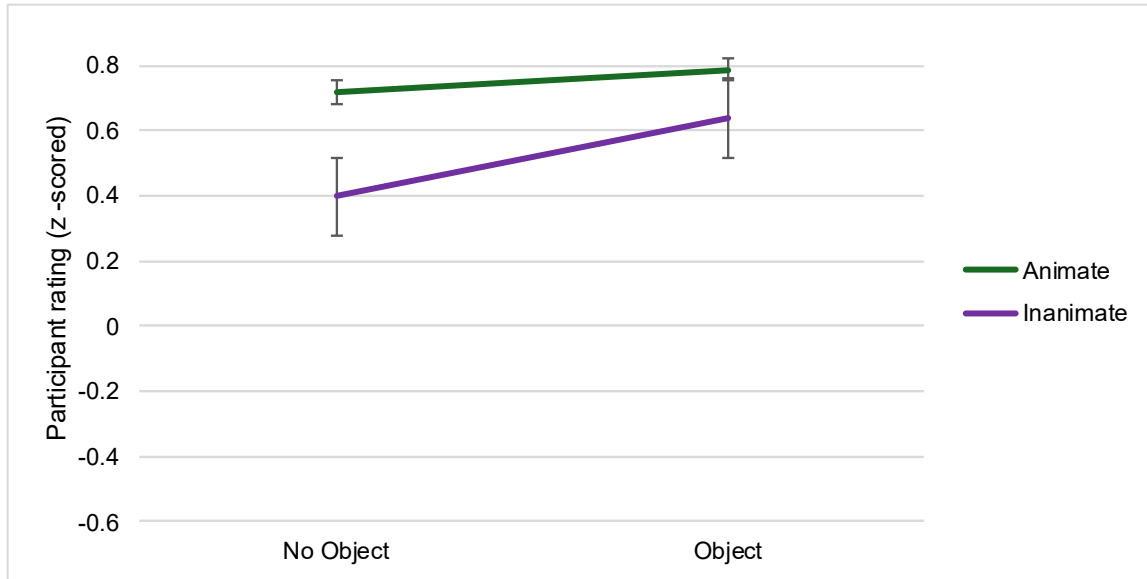
**Table 4:** Latin square design for Experiment 2

#### 4.4 Predictions

The predictions for Experiment 2 were the same as those for Experiment 1. As before, it tested the hypothesis that an interaction would be found between the factors of animacy and object presence. As will be seen, this hypothesis was confirmed by the results.

#### 4.5 Analysis

Figure 2 reveals results that parallel those found from Experiment 1. One difference that may be observed is that the acceptability ratings are higher overall than in the previous experiment; this may reflect the design improvements made in Experiment 2. However, ratings for object omission with inanimate subjects remain lower relative to corresponding sentences with objects than ratings for object omission with animate subjects. Moreover, while differences were observed between different verbs (see Appendix 2), there was no dichotomous pattern that might seem to reflect a distinction between verbs with and without lexicalized distinctions. These findings confirm the results of the previous experiment and provide further support for the predictions regarding the effect of animacy and object omission upon acceptability.



**Figure 2:** Participant ratings by condition (Experiment 2)

The same analysis was conducted for Experiment 2 as for Experiment 1. The model structure, effects, and coding were the same as in Experiment 1. Regression formulae and other analytical code for this experiment are also available in the Supplementary Files.

The regression analyses again found significant results. In the mixed-effects linear regression, animacy was significant as a main effect ( $\beta=0.32$ ,  $t=6.83$ ,  $p<0.001$ ), as was object presence ( $\beta=0.24$ ,  $t=5.67$ ,  $p<0.001$ ); the interaction between the two was also significant ( $\beta=-0.17$ ,  $t=-3.87$ ,  $p<0.001$ ). It is striking that although the effect sizes are smaller than in Experiment 1, as also shown in Figures 1 and 2, the significance levels remain similar. These results provide further confirmation for the theories discussed in Section 2, and are inconsistent with the null hypothesis that no interaction between animacy and object presence would exist.

#### 4.6 Discussion

Experiments 1 and 2 successfully established the existence of an interaction between animacy and object omission. These findings provide confirmation for the preliminary observations that have been made by a number of previous authors, as described in Section 2. Moreover, this interaction has been found using a dataset with multiple verbs and scenarios. Although conventionalization was not explicitly encoded as a variable in these experiments (see also Section 6), at least some of these scenarios involve actions that are far from conventionalized, as shown in Table 4 (e.g. *pushing boats*, *pushing boxes*). However, while the relevance of animacy as a factor has been established, questions remain to be addressed regarding the nature of the phenomena responsible for this interaction, given that animate and inanimate subjects differ in multiple respects. As described in Section 2, several authors (e.g. Martí

2015; Rappaport Hovav & Levin 2024; Folli & Harley forthcoming) have suggested that intentionality is a key property of animate subjects. Accordingly, Experiment 3 examined the potential role of intentionality as a factor.

## 5 Experiment 3

In its design and protocol, Experiment 3 broadly followed Experiment 2. However, the aim of Experiment 3 was not to compare sentences with animate and inanimate subjects, but rather to compare sentences with animate subjects in control of their actions (i.e. acting intentionally) and with animate subjects not in control of their actions (i.e. acting unintentionally). As before, the experiment was preregistered prior to data collection.<sup>11</sup>

### 5.1 Participants

Experiment 3, like Experiment 2, had a recruitment target of 180 participants. The inclusion criteria were the same as for Experiments 1 and 2; participants from the first two experiments were not eligible for Experiment 3. A total of 229 participants were recruited, of whom 47 were excluded for poor performance on control questions and 2 chose to withdraw after completing the study. As described in Section 5.2, this increase in the exclusion rate may have been due to a change in the control sentences used. Participant recruitment was again conducted online using Prolific, with the same compensation rates as before.

### 5.2 Materials and procedure

In Experiment 3, test sentences and procedures followed the same format as in Experiment 2. Four stimulus sets were constructed for each verb; however, the list of verbs was changed slightly (see Table 5), owing to the difficulty with some verbs of constructing a sufficient number of scenarios in which the subject could plausibly be described as not in control of the action (e.g. *accidentally clean/bring*). The format and nature of control items was the same as in the previous experiment; as in Experiment 2, participants were excluded if they achieved less than 80% accuracy on all control sentences combined. Participants seemed to have an unexpectedly high tolerance for object omission with *remove*, as compared with *make*; across all participants, sentences in which *remove* was used without an object (see 20) received positive ratings in 34% of cases, as opposed to only 5% for sentences in which *make* was used without an object. However, enough participants met the inclusion criteria in spite of this phenomenon that *remove* was retained as a control verb.

- (20) Intentional (i.e. best-case) object-omission stimuli with *remove*
- a. Mary got a trowel and carefully removed some soil from the flowerbed.

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<sup>11</sup> [https://osf.io/jr8bx/overview?view\\_only=682e8900238b476c9645ba903119f832](https://osf.io/jr8bx/overview?view_only=682e8900238b476c9645ba903119f832)

She removed very gently.

- b. Mary reached over and carefully removed the drive from her computer.

She removed very gently.

- c. Mary came in and carefully removed her coat.

She removed very slowly.

- d. Mary got a knife and carefully removed some peel from an apple.

She removed very slowly.

| Verb   | Type                        |
|--------|-----------------------------|
| pull   | Test verb                   |
| push   | Test verb                   |
| scrape | Test verb                   |
| splash | Test verb                   |
| squish | Test verb                   |
| wipe   | Test verb                   |
| make   | Control verb (transitive)   |
| remove | Control verb (transitive)   |
| go     | Control verb (intransitive) |
| laugh  | Control verb (intransitive) |

**Table 5:** Verbs used in Experiment 3

### 5.3 Design

The design of Experiment 3 was essentially the same as that of Experiment 2. The main difference was that the independent variables were object presence and in-control status, rather than animacy. A sample stimulus set can be seen in Table 6. As this table shows, another difference is that in-control and out-of-control sentences do not use the same adverbs; it was found that few adverbs seemed equally compatible with both scenarios, and so this change was made to minimize the potentially confounding effects of infelicity. Test sentences were also in the simple past rather than the present progressive, owing to the speed at which most out-of-control events would take place (e.g. *#John is accidentally pushing the door open*). In other respects, the design and the distribution of test and control items remained the same as in Experiment 2.

|        | Group A                                                                                      | Group B                                                                              | Group C                                                                              | Group D                                                                      |
|--------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Item 1 | John tripped and accidentally pushed a box down the ramp. He pushed the box unintentionally. | John came and impatiently pushed a box down the ramp. He pushed the box really hard. | John tripped and accidentally pushed a box down the ramp. He pushed unintentionally. | John came and impatiently pushed a box down the ramp. He pushed really hard. |
| Item 2 | John came out and carefully pushed Mary's car out of                                         | John reversed without looking and accidentally pushed                                | John came out and carefully pushed Mary's car out of                                 | John reversed without looking and accidentally pushed                        |

|        |                                                                                         |                                                                                        |                                                                                |                                                                               |
|--------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|        | the drive. He pushed the car really hard.                                               | Mary's car out of the drive. He pushed the car unintentionally.                        | the drive. He pushed really hard.                                              | Mary's car out of the drive. He pushed unintentionally.                       |
| Item 3 | John tripped and accidentally pushed the door open. He pushed the door unintentionally. | John walked over and carefully pushed the door open. He pushed the door really hard.   | John tripped and accidentally pushed the door open. He pushed unintentionally. | John walked over and carefully pushed the door open. He pushed really hard.   |
| Item 4 | John walked up and impatiently pushed the doorbell. He pushed the bell really hard.     | John slipped and accidentally pushed the doorbell. He pushed the bell unintentionally. | John walked up and impatiently pushed the doorbell. He pushed really hard.     | John slipped and accidentally pushed the doorbell. He pushed unintentionally. |

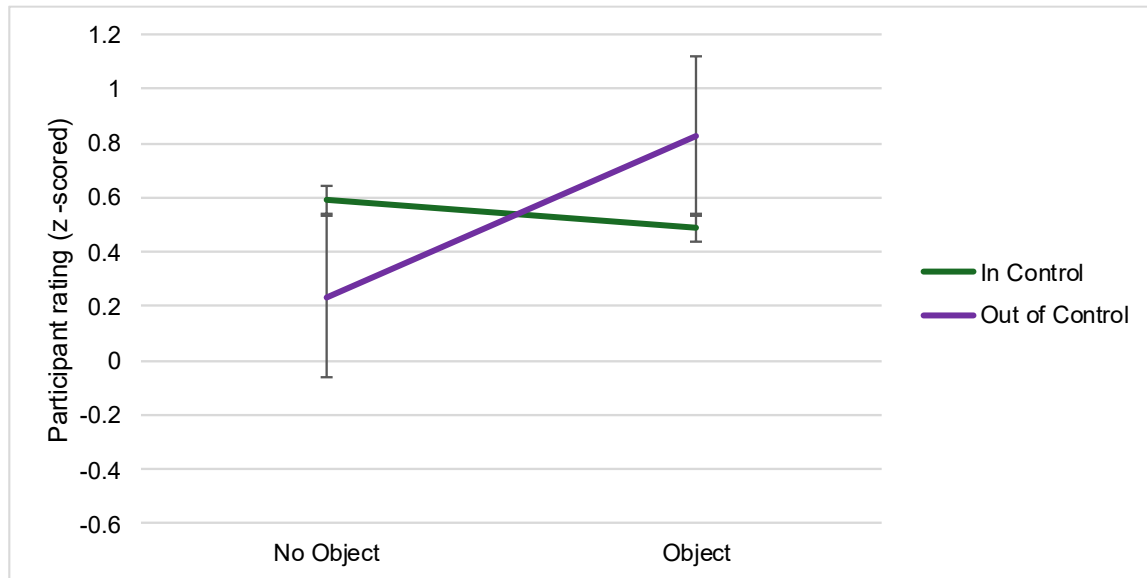
**Table 6:** Latin square design for Experiment 3

#### 5.4 Predictions

Experiment 3 tested the prediction that an interaction would be found between the factors of in-control status and object presence. The null hypothesis was that no such interaction would be found. No prediction was made as to whether in-control status and object presence would be significant in their own right as main effects, since this is a question beyond the scope of the theories discussed in Section 2. It will be seen that the results of this experiment confirm the prediction made.

#### 5.5 Analysis

Figure 3 displays the results of this experiment. As can be seen, the out-of-control objectless condition has the lowest acceptability rating. Whereas in the previous experiments, sentences with animate subjects scored consistently higher than inanimate subjects even in the overt-object condition, here in the overt-object condition animate subjects not in control of their actions seem to be preferred to subjects in control of their actions. However, this preference may more plausibly be ascribed to the different adverbial phrases used for the two conditions than to some preference for unintentional action per se. As before, some variation was observed among different verbs (see Appendix 2).



**Figure 3:** Participant ratings by condition (Experiment 3)

The same analysis was conducted for Experiment 3 as for Experiment 2. The only change was that the independent variable of animacy was replaced by in-control status, which was coded in the same way. Regression formulae and other analytical code for this experiment are also available in the Supplementary Files.

The regression analyses again found significant results. In the mixed-effects linear regression, in-control status was significant as a main effect ( $\beta=0.36$ ,  $t=4.87$ ,  $p<0.001$ ), as was object presence ( $\beta=0.59$ ,  $t=8.79$ ,  $p<0.001$ ); the interaction between the two was also significant ( $\beta=-0.70$ ,  $t=-9.94$ ,  $p<0.001$ ). These results support the view that intentionality, represented as in-control status, is a salient factor in the association between animacy and object omission, and are inconsistent with the null hypothesis that no interaction between in-control status and object presence would exist.

## 5.6 Discussion

The aim of this experiment was to test the hypothesis that an interaction would exist between object omission and intentionality, similar to that found for object omission and animacy. This hypothesis was confirmed, just as the animacy hypothesis was confirmed by the previous experiments. As will be seen in Section 6, these findings have implications for theories of object omission, although some outstanding questions remain.

## 6 Discussion

This study investigated the research question of how and why animacy exercises a significant effect upon the acceptability of object omission. It was predicted that acceptability judgement ratings would show an interaction between the factors of animacy and object omission. In addition, the hypothesis that these animacy effects were related to intentionality predicted that

a similar interaction would exist when in-control status for animate subjects was included as a factor. As shown above, the results of these experiments support these predictions. While such effects have been predicted by several different theories, it will be seen there are other respects in which these theories make different predictions. Future research may be able to address these predictions and assist in the comparative evaluation of the underpinning theories. The findings of this study also raise broader questions regarding object omission and the role of animacy within grammar.

### **6.1 Previous theories**

As described in Section 2, the interaction of object omission and animacy/intentionality is predicted by several different theories, based upon very different assumptions. For Martí (2015), animacy effects are a sign of the incorporation of a null noun in object omission, reflecting a possible cross-linguistic constraint upon noun incorporation more generally. For Rappaport Hovav & Levin (2024), animacy effects reflect lexically specified restrictions on thematic roles. For Folli & Harley (forthcoming), animacy restrictions provide the sortal criteria needed for the interpretation of unsaturated predicates. While all these theories, in one way or another, predict the existence of interactions such as those found in the present study, they make different predictions regarding the specific properties responsible for such interactions and regarding their lexical distribution.

One question raised by previous work is the exact nature of the property or properties responsible for animacy-related restrictions. As described in Section 2, many previous theories have invoked volition/intentionality as a key property of animate subjects. In some accounts (e.g. Martí 2015; Folli & Harley forthcoming), this is said to reflect a pervasive grammatical phenomenon. Although intentionality is also considered crucial by Rappaport Hovav & Levin (2024: 7), as their analysis focuses on a single verb, less discussion is made of what role intentionality might play elsewhere. It might be expected that in a lexical account, there would be no necessity for the encoding of intentionality to pattern systematically with the encoding of object omission; however, their analysis does not explicitly claim that intentionality is lexically encoded in the same way proposed for object omission. The concept of intentionality can itself be decomposed further. In a study focusing on truth-value judgements rather than grammaticality judgements, Joo et al. (2025) found that the most relevant factor was whether subjects were in control of their physical actions, rather than whether they were acting in pursuit of a particular goal. In the present study, these two components of intentionality were not manipulated separately, being either both present or

both absent; however, further research may reveal whether these different components impact grammaticality judgements in the same way as truth-value judgements.

Another factor proposed as facilitating object omission is conventionalization. In accounts such as Goldberg (2001), Glass (2014, 2022) and Rappaport Hovav & Levin (2024) it is suggested that object omission is facilitated by the routine, conventionalized nature of the event denoted, with or without a specific animacy restriction as well. Such accounts would predict that object omission with animate subjects performing conventionalized activities would be more acceptable than with animate subjects performing novel activities. Conversely, an account such as Folli & Harley (forthcoming) would predict that, given an appropriate context, object omission should be equally available for more and for less conventionalized activities. These predictions intersect with predictions made regarding the role of the lexicon in object omission. As described above, the experiments presented here identified animacy effects using a dataset including at least some less conventionalized scenarios; however, conventionalization was not tested systematically as a factor. Further work is needed to identify the extent to which conventionalization per se plays a role in facilitating object omission.

The question also remains of the extent to which object omission is lexically encoded. As described in Section 2, accounts such as Rappaport Hovav & Levin (2024) interpret differences such as those in (4) as the result of separate lexical entries for “*basic-sweep*” and “*broom-sweep*”, while other accounts (e.g. Rice 1988; Folli & Harley forthcoming) consider it more parsimonious to interpret object omission as a pervasive and productive grammatical phenomenon, and to seek semantic explanations for the variation observed. Accounts of the former type predict consistency for individual lexemes and potentially wide variation between lexemes, while accounts of the latter type predict a more consistent pattern across lexemes. The experiments presented here did identify some differences in behaviour among different verbs (see Appendix 2); however, the small number of verbs and the variation that also exists among individual items makes it difficult to interpret such variation. Further work remains to be done in order to investigate the full extent of lexical variation in object omission and its availability even in nonce uses of such constructions.

## **6.2 Object omission and animacy as independent factors**

One noteworthy finding of these experiments is the significance of object omission as a factor in its own right, independent of animacy. This in itself is not unexpected, given the widely observed variation among constructions in whether they allow object omission. However, little previous work has addressed the specific question of whether object omission might

tend to lower acceptability ratings even with volitional, animate subjects, and even when lexical, syntactic, and contextual factors were held constant. As stated above, this study made no predictions as to whether such an effect would be found. The existence of such an effect affirms the gradient approach to the acceptability of object omission taken by authors such as Glass (2022), although Glass’s study did not directly compare sentences with and without overt objects.

The significance of animacy as a factor, even with overt objects, also deserves comment. Although it has long been recognized that subjects with characteristically animate properties such as volition and sentience are somehow more “prototypical” than subjects without these properties (e.g. Dowty 1991), it might not have been predicted that simply changing the subject of a sentence with an overt object from animate to inanimate would produce a significant decrease in its acceptability rating, given that all sentences used were constructed by a native speaker with the goal of equal acceptability. Again, no prediction was made in this study as to whether such an effect would exist. That it was found exemplifies the pervasive influence of animacy-related properties upon linguistic phenomena (cf. Folli & Harley 2008). No such decrease was found for sentences with animate subjects not in control of their actions; if anything, such sentences were preferred. However, as discussed in Section 5, the methodological differences between experiments make it difficult to compare their results directly in this respect.

## **7 Conclusion**

The aim of this study was to investigate the role of subject animacy and intentionality upon the acceptability of object omission. A series of three experiments confirmed the hypotheses that acceptability judgement ratings would show an interaction between the factors of animacy and object omission, and between the factors of in-control status and object omission. All of these factors were also found to be significant in themselves. These findings support the predictions made by several theories of object omission in English, and provide robust quantitative data to substantiate a phenomenon whose existence has previously been noted mainly in passing, namely the relevance of animacy of the external argument in the availability of object omission in English. Furthermore, the study shows that intentionality is a crucial property of animate subjects in facilitating object omission. However, further work is needed to provide a comprehensive categorisation of all the factors involved in object omission, and to test additional competing predictions made by different accounts.

As described in Section 6.1, previous analyses of object omission have made a number of predictions that were not tested directly in the present study. The inclusion of

conventionalization as a variable in a study of similar design could provide data as to whether this is also a factor favouring the use of animate subjects with object omission. Similarly, the inclusion of a greater number of verbs, with more items per verb, could clarify the extent to which subject animacy and object optionality are encoded at the lexical level. Further work may also identify additional factors responsible for some of the variation identified in the present study within individual lexemes. Research on these themes has the potential to contribute to broader questions regarding the mechanisms by which grammatical distinctions are encoded.

**Supplementary files**

All data associated with this study are available at

[https://osf.io/8mquw/overview?view\\_only=057323e662184c1ab2f1ff1c81e39ec2](https://osf.io/8mquw/overview?view_only=057323e662184c1ab2f1ff1c81e39ec2).

**Ethics and consent**

Ethical approval for this study was granted by [ANONYMIZED]. Informed consent was obtained from all participants.

**Funding information**

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**Competing interests**

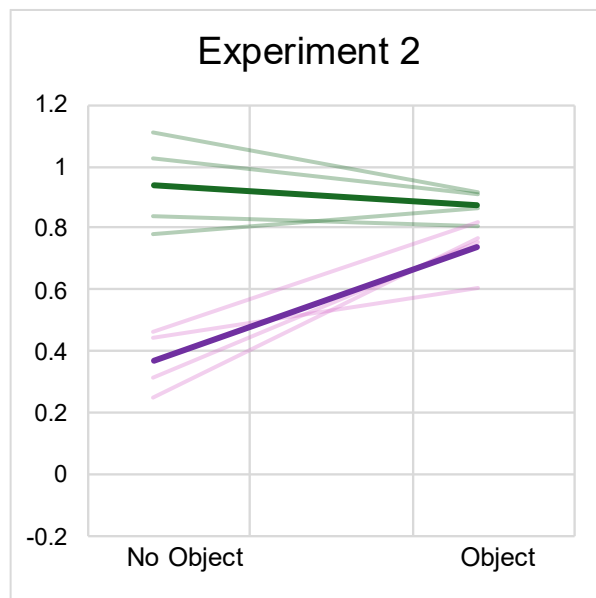
The authors have no competing interests to declare.

## Appendix 1

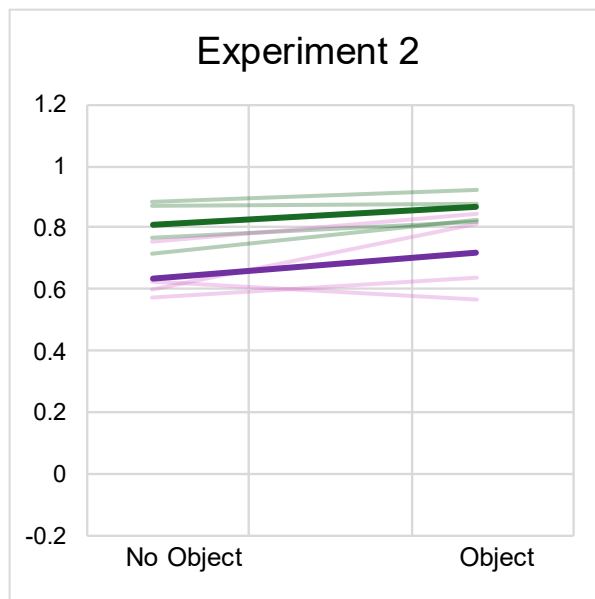
| Experiment | Factor                  | $\beta$ | $z$    | $p$    |
|------------|-------------------------|---------|--------|--------|
| 1          | Animacy                 | 2.34    | 5.91   | <0.001 |
|            | Object                  | 2.58    | 5.69   | <0.001 |
|            | Animacy * Object        | -0.98   | -2.33  | 0.02   |
| 2          | Animacy                 | 1.39    | 7.44   | <0.001 |
|            | Object                  | 1.48    | 4.64   | <0.001 |
|            | Animacy * Object        | -0.44   | -2.51  | 0.01   |
| 3          | Intentionality          | 1.36    | 4.89   | <0.001 |
|            | Object                  | 2.82    | 7.24   | <0.001 |
|            | Intentionality * Object | -3.05   | -10.79 | <0.001 |

**Table 7:** Cumulative Link Mixed Model Regression Results

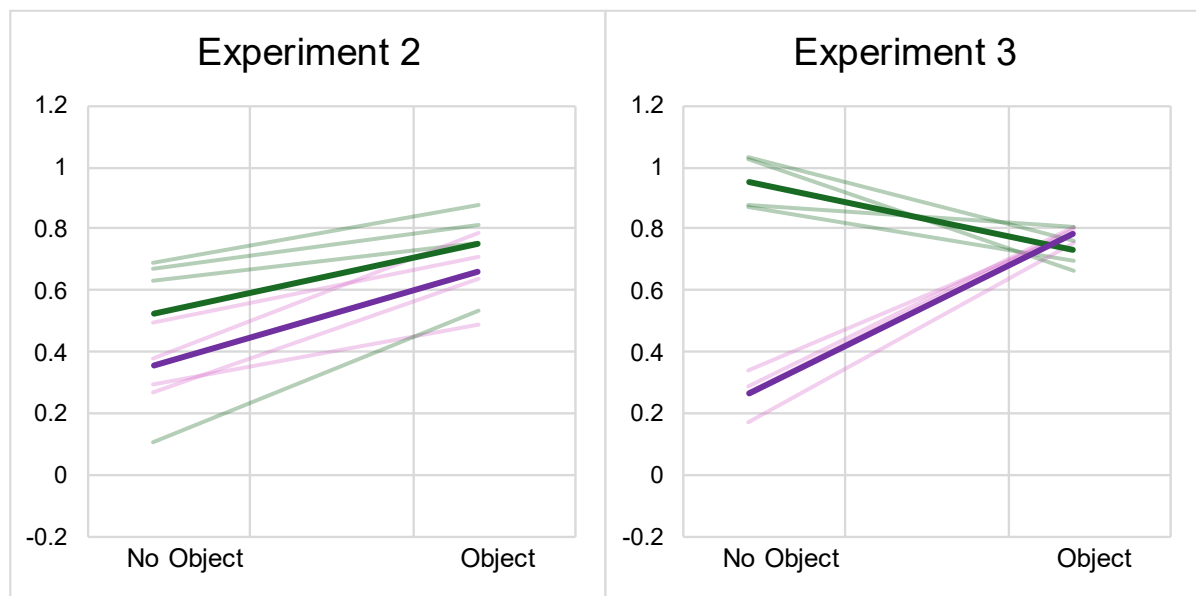
## Appendix 2



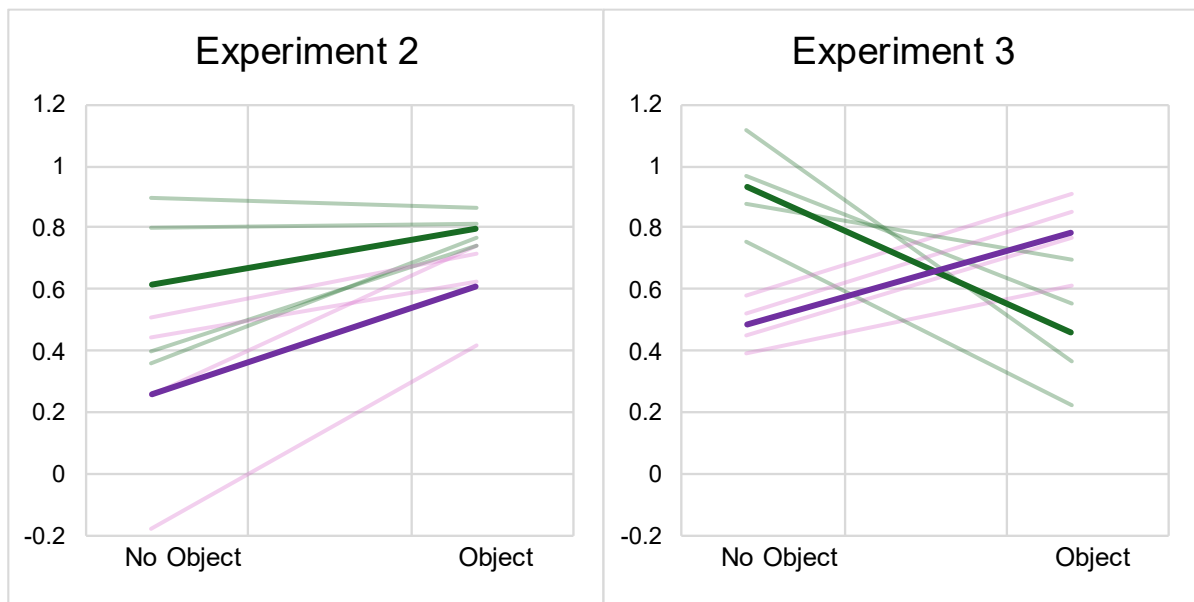
**Figure 4:** Participant ratings for *cook*, showing mean scores for all items combined (bold) and for individual items



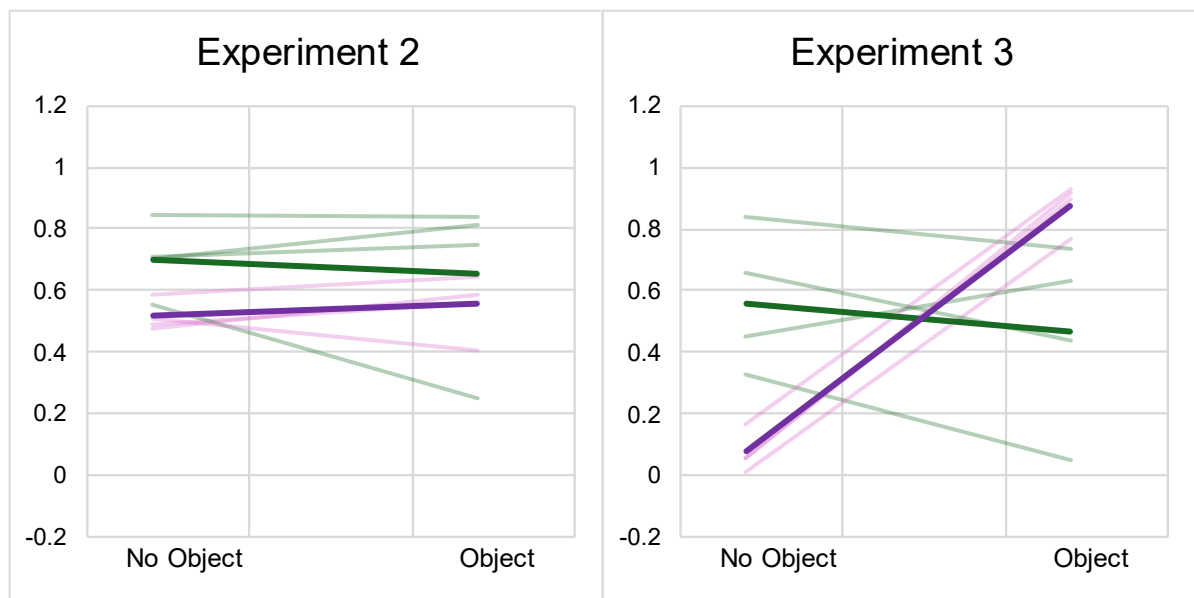
**Figure 5:** Participant ratings for *clean*



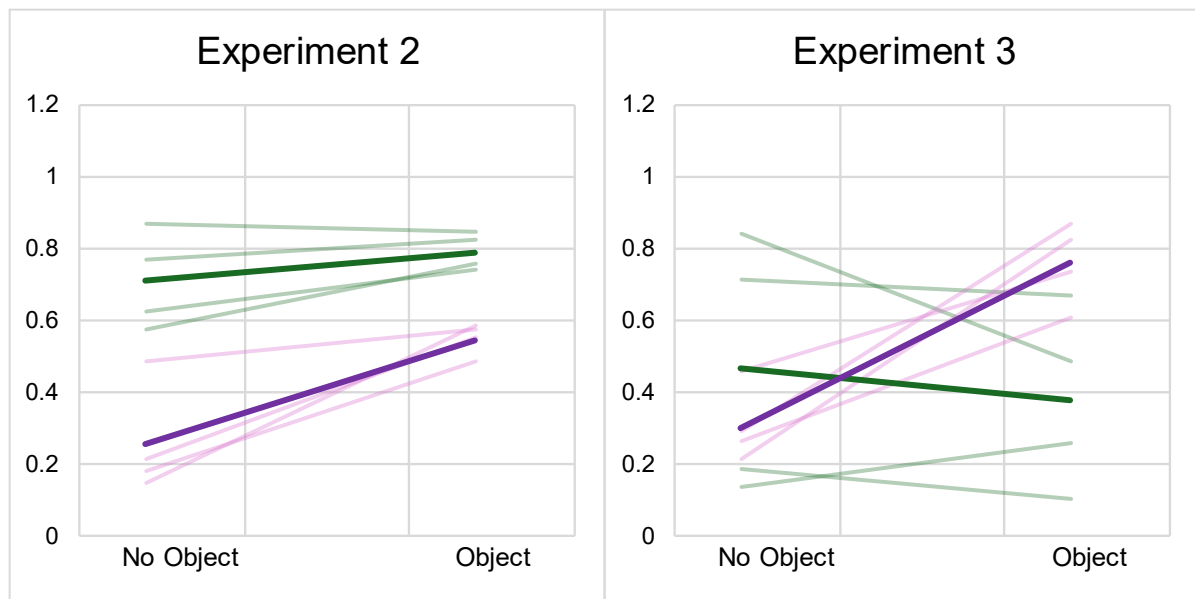
**Figure 6:** Participant ratings for *pull*



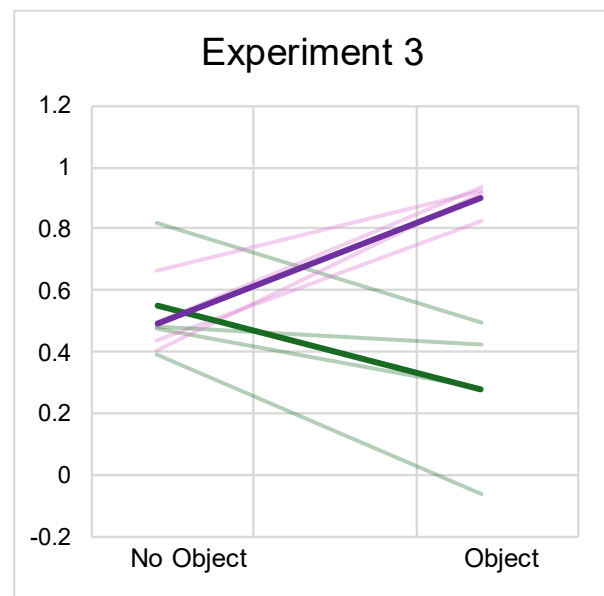
**Figure 7:** Participant ratings for *push*



**Figure 8:** Participant ratings for *scrape*



**Figure 9:** Participant ratings for *wipe*



**Figure 10:** Participant ratings for *splash*



**Figure 11:** Participant ratings for *squish*

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